App Control

Note: Before using the mobile app for control, it's recommended to first go through "8. Remote Tool Installation and Container Access → 8.1 VNC Installation and Connection" and "Robot Version Configuration Instruction". Make sure the robot is set to the correct hardware configuration before starting control via the app.

1. App Installation and Connection

You can use the "WonderPi" app to control MentorPi. Follow the steps below to install and connect the app.

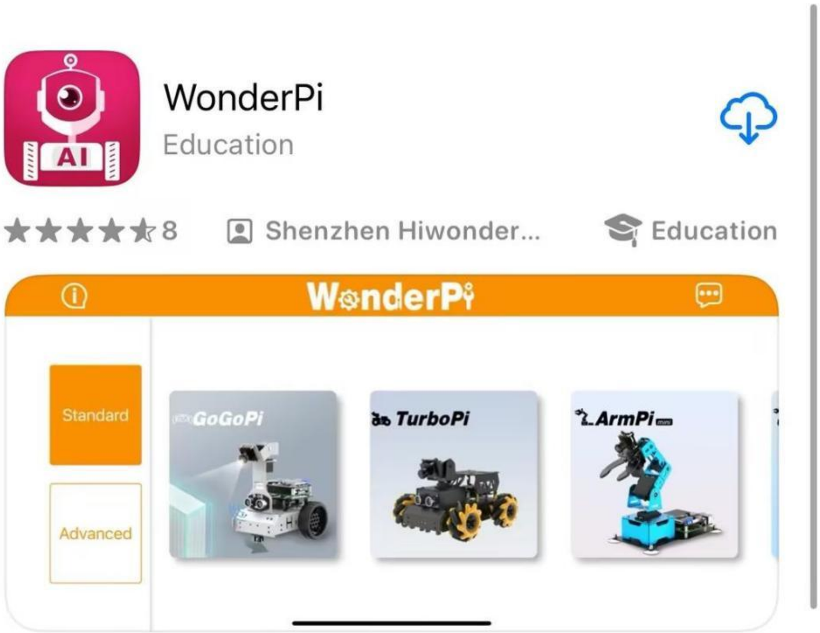
Note:

① Please ensure that all permissions are granted to the app during installation to ensure its proper functionality.

② Before launching the app, make sure that both GPS and Wi-Fi are enabled on your phone.

1.1 App Installation

1) **For Android users**: Locate the app installation package in the provided directory, then transfer it to your smartphone and install it manually.
2) **For iOS users**: Download **"WonderPi"** directly from the **App Store**.



1.2 Connection Mode

After installing the app, connect the robot. Here are two network modes for the robot:

①**AP direct connection mode:** the controller can turn on the hotspot to connect with the phone, but it cannot connect to an external network.

②**STA LAN mode:** the controller can actively connect to a designated hotspot or Wi-Fi, enabling communication with external networks.

The robot is the AP direct connection mode by default. No matter which mode is selected, you can achieve the same game functions of the MentorPi app.

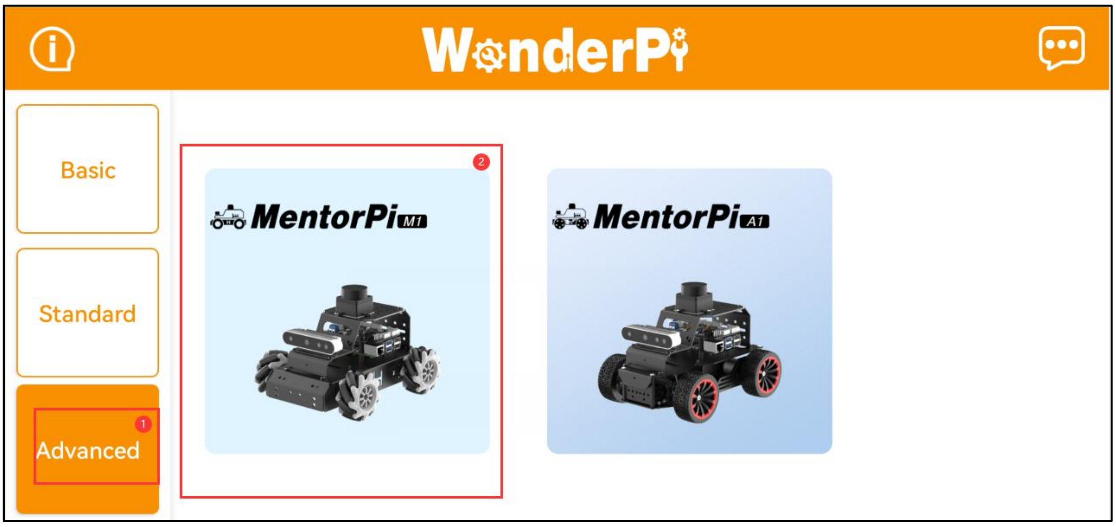
It is recommended to learn the AP direct connection mode to experience the corresponding functions first. The LAN mode can be viewed based on your own needs.

1.2.1 Direct Connection Mode (Must Read)

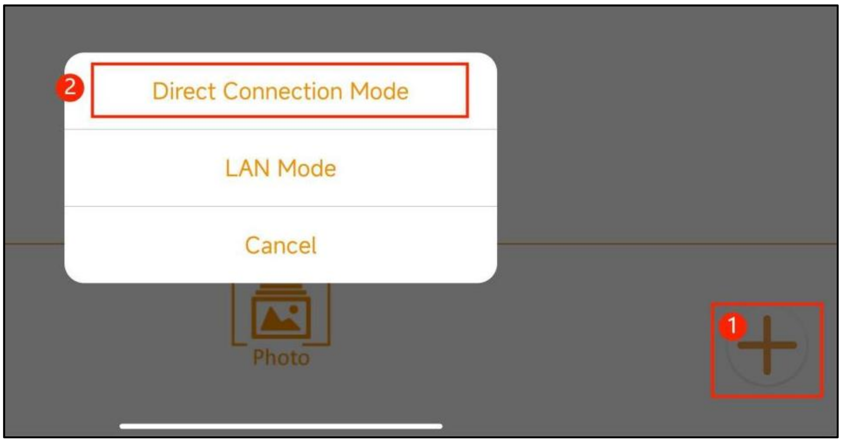
Connecting to the MentorPi M1 is used as an example; the same instructions also apply to the MentorPi A1.

1) Open the **WonderPi** app and select **Advanced > MentorPi-M1** in sequence.

Note: If you are using the Ackermann chassis version, select MentorPi-A1 instead.

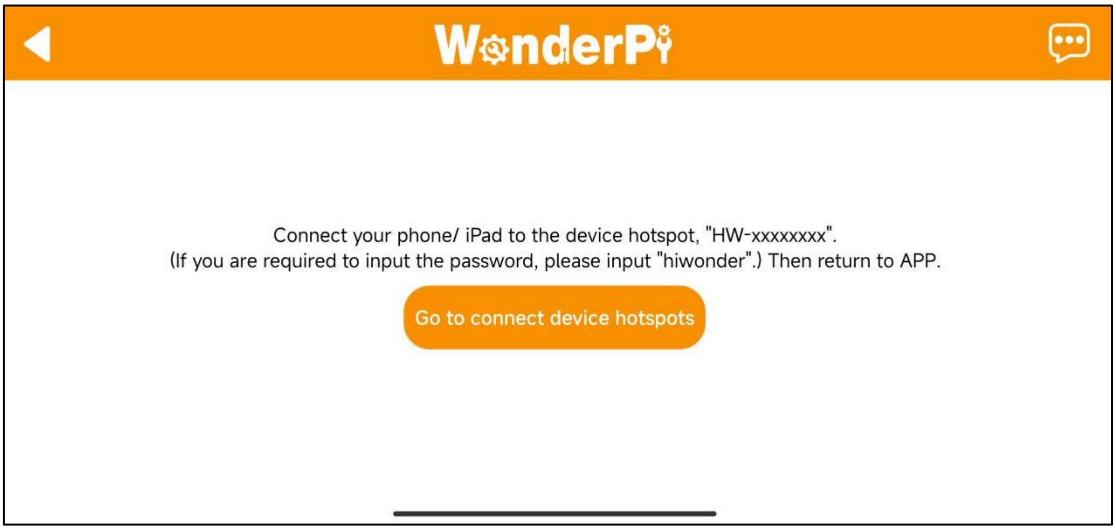


2) Click the "+" button at the lower right corner to select direct connection mode.

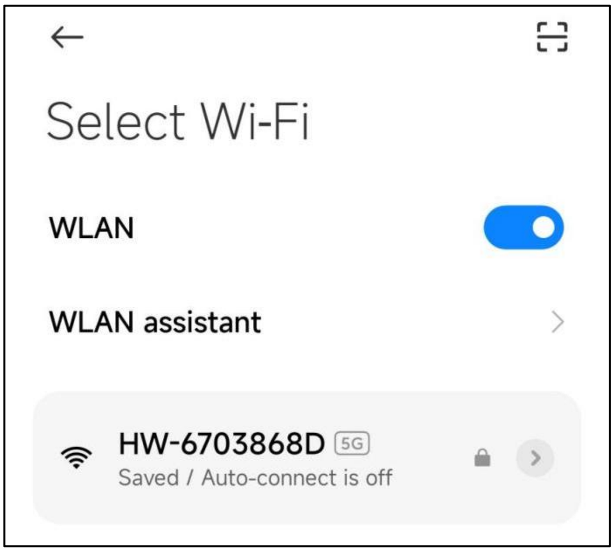


Note: If you use the LAN mode, please refer to the section "**1.2.2 LAN mode connection**".

3) Tap **"Go to connect device hotspots"** to open the settings page, and connect to the Wi-Fi hotspot broadcasted by **MentorPi**.



4) When prompted, enter the password: **hiwonder**.



Note (for iOS users):

After connecting to the hotspot, wait until the message **"Wi-Fi connected successfully"** appears before returning to the app. If you return too early, the device may not be detected. In that case, tap the **Refresh** button  several times to search for the device again.

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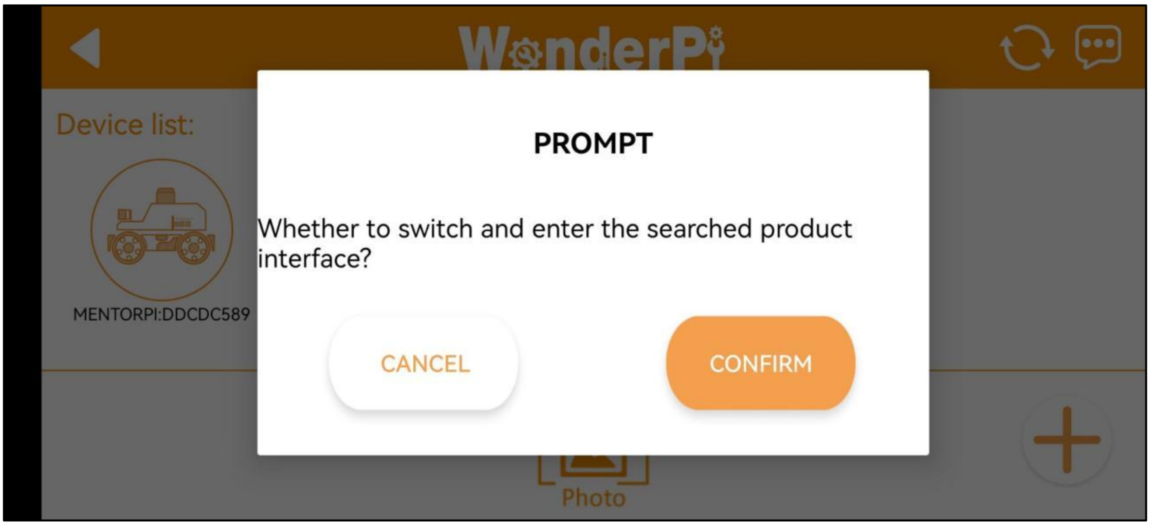
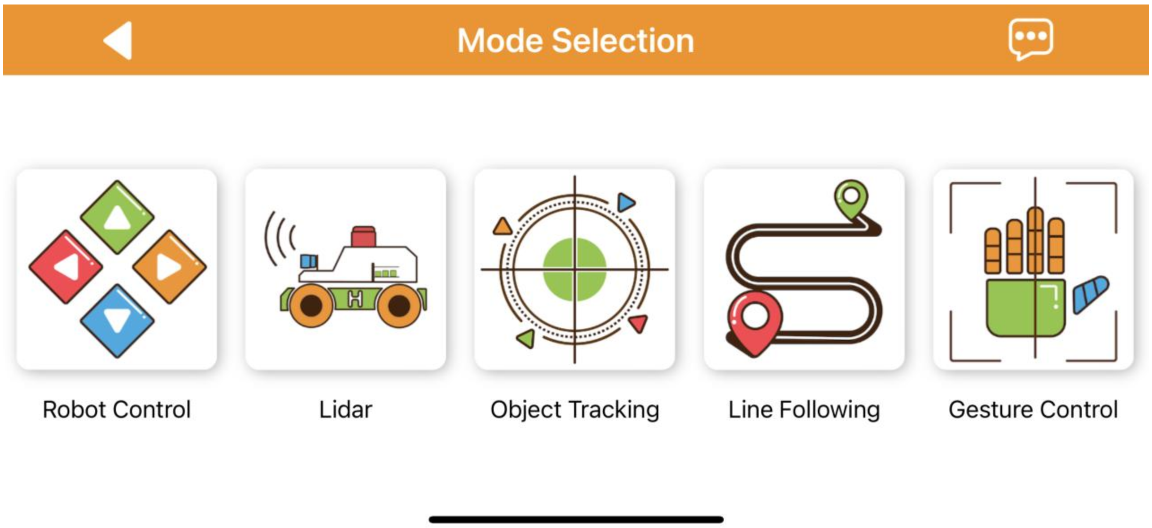
5) Once connected, return to the app. The phone will automatically establish a connection with the device. When the robotic arm icon appears, it indicates that the connection has been successfully established.

Note: If a pop-up message appears stating **"Network unavailable. Continue?"**, tap **"Keep Connection"** to proceed.

6) If the message **"Whether to switch and enter the searched product interface?"** appears, it means the selected product version in **Step 1** was incorrect. Tap **"OK"** to automatically switch to the correct version selection screen.



7) The game mode selection interface will then appear, as shown below.



For detailed instructions on how to operate each game mode, please refer to **Section 2: App Games Introduction**.

1.2.2 LAN Mode Connection

Note:

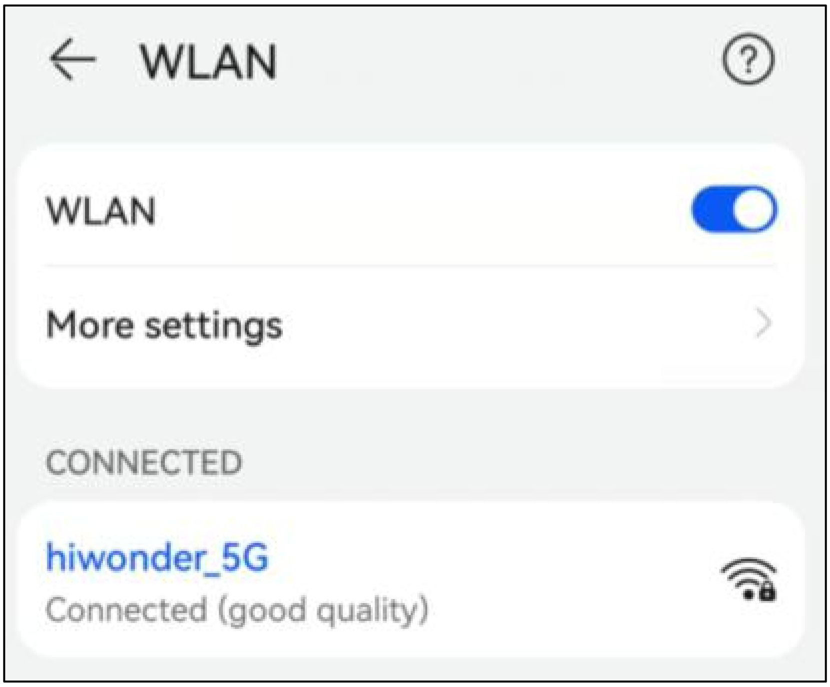
① When MentorPi starts up in LAN mode, it will first attempt to automatically connect to a preset network. During this search, LED2 on the Raspberry Pi expansion board will blink rapidly, indicating that it is scanning for the target network. If the network is not found after three search attempts, MentorPi will automatically switch to Direct Connection mode, indicated by LED2 blinking slowly. At this point, you can connect to MentorPi's built-in hotspot.

② If MentorPi is set to connect to a specific device's hotspot in LAN mode, but that device becomes temporarily unavailable, you can use another device instead. Simply configure the new device to use the same hotspot name and password as the original one—MentorPi will be able to connect to it automatically.

③ When operating in LAN mode, MentorPi does not broadcast its own hotspot.

1) Connect your phone to a **5G Wi-Fi network**. For example, connect to **"Hiwonder_5G"**.

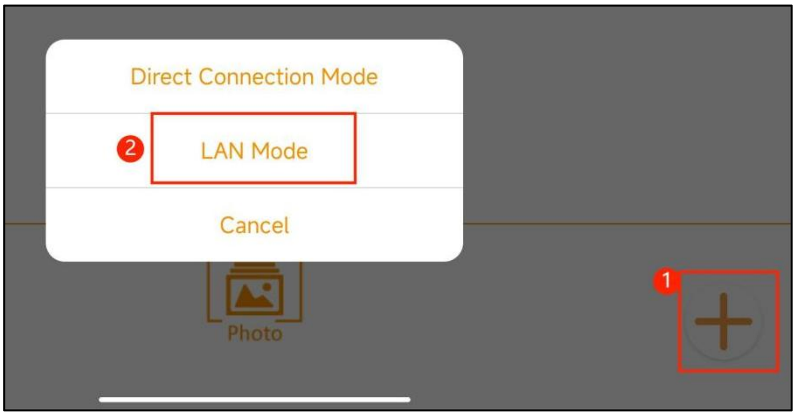
If your router supports both 2.4G and 5G bands and they are configured with separate SSIDs, the Wi-Fi names will typically reflect this—for instance, **"Hiwonder"** for 2.4G and **"Hiwonder_5G"** for 5G.



2) Open the **WonderPi** app and, select **Advanced > MentorPi-M1**.

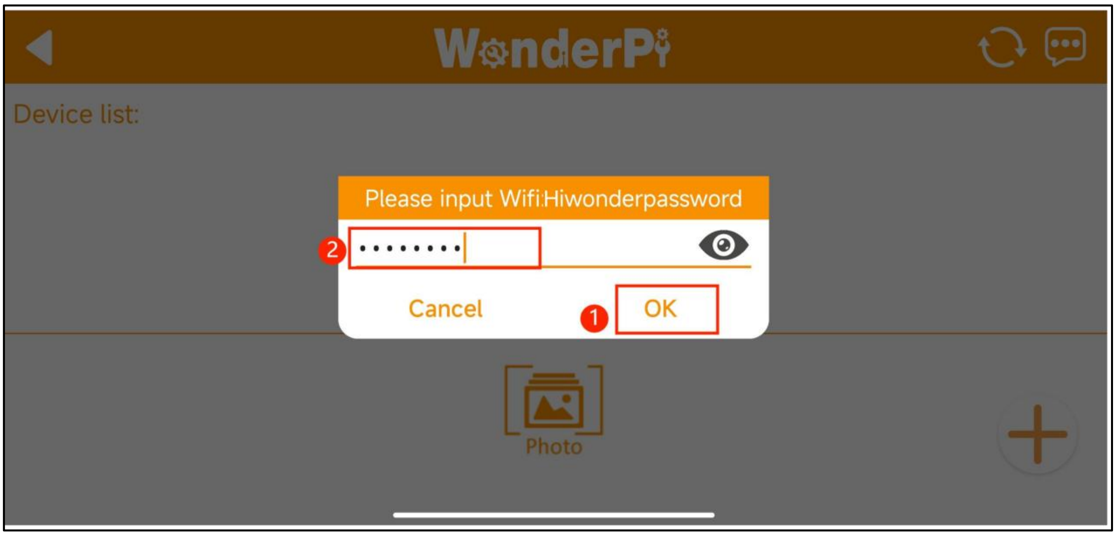
Note: If you're using the Ackermann chassis version, select **MentorPi-A1**.

3) Tap the **"+"** button in the lower-right corner and choose **LAN Mode**.

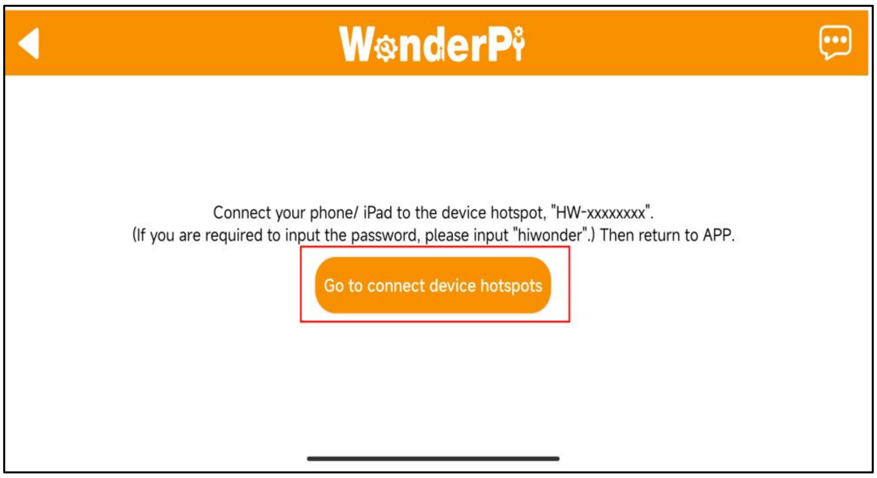


4) The app will prompt you to enter the Wi-Fi password.

Please make sure to enter the correct password—otherwise, the connection will fail. Once entered, tap **"OK"**.



5) Tap “Go to connect device hotspots”.

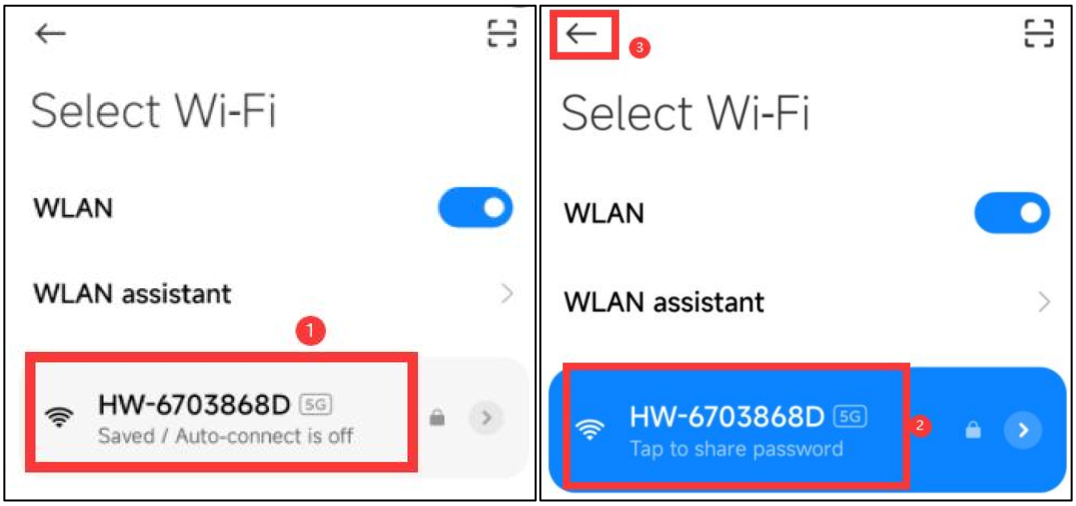


6) Your phone will automatically open the Wi-Fi settings page.

Look for a hotspot beginning with **"HW"**, and connect using the password:

hiwonder.

Once connected, return to the app.



7) The app will then begin **automatic network configuration**.

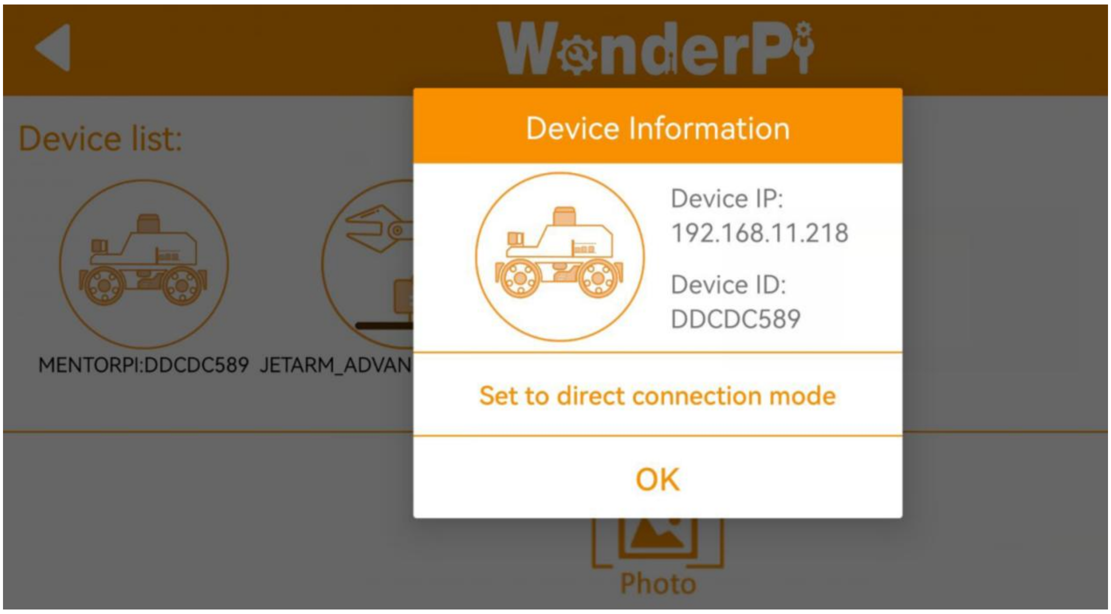
8) After a short while, the main interface will display the **MentorPi** icon, and the **LED light** on the expansion board will remain steady—indicating a successful connection.

9) Long-press the MentorPi icon in the app to view its assigned **IP address** and **device ID**.

10) You can use this IP address with a remote desktop tool to access the device.

For detailed steps, please refer to **Section 8: Set Up Development Environment**.

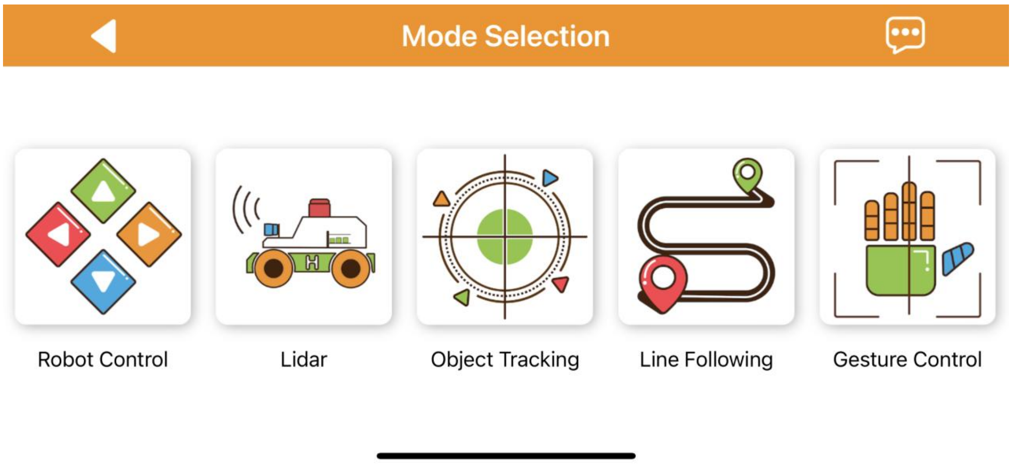
11) To switch from **LAN mode** back to **Direct Connection mode**, press and hold the **KEY1** button on the expansion board until the **blue LED flashes**, indicating the mode has been switched successfully.



2. App Games Introduction

In this section, we'll use the **iOS system** for demonstration purposes. The instructions also apply to **Android devices**.

The app offers **five game modes**: **Robot Control**, **Lidar**, **Object Tracking**, **Line Following**, and **Gesture Control**.



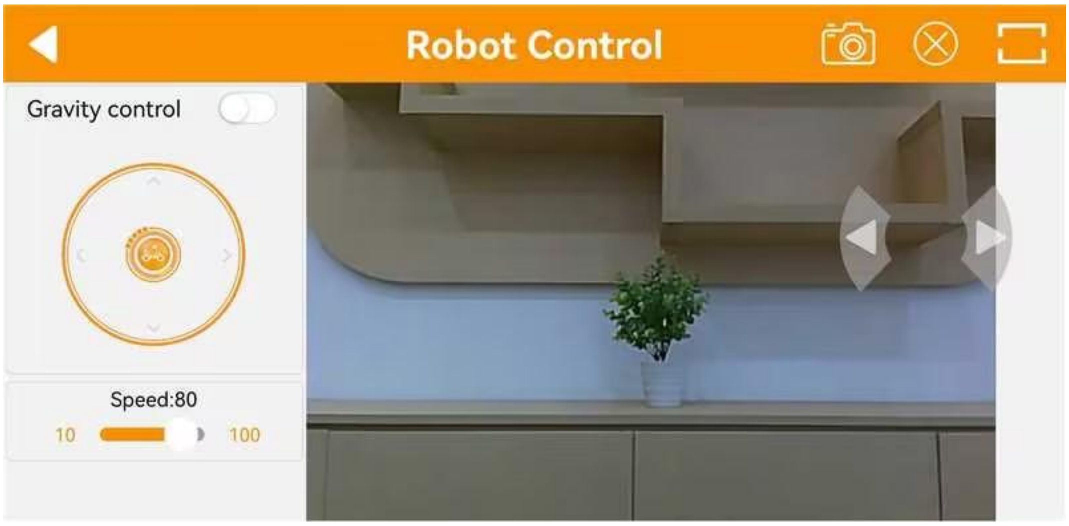
The following table provides a brief description of each game mode:

Icon	Game Mode	Function
	Robot control	Manually control the movement and direction of the robot
	Lidar	Includes three modes: Obstacle Avoidance, Lidar Following, and Lidar Guarding
	Object tracking	Select a target object color; the robot will automatically track the selected object
	Line following	Place a line on the ground and set its color as the recognition target; the robot will follow the line path
	Gesture control	(Available only on the Mecanum wheel version) Use hand gestures to control the robot's movement direction based on finger movement tracking.

2.1 Robot Control

Tap "Robot Control" on the mode selection screen to enter the game's control interface. Note that the interface may vary slightly between the Mecanum wheel and Ackermann wheel versions.

1) The robot control interface for the mecanum wheel version is as below:



①**Left Controls**:

From top to bottom — toggle gravity sensing, control forward/backward movement, and adjust speed.

②**Center Display**:

Displays the live camera feed. You can drag the screen to adjust the camera's pan-tilt angle (available only on the 2D pan-tilt version).

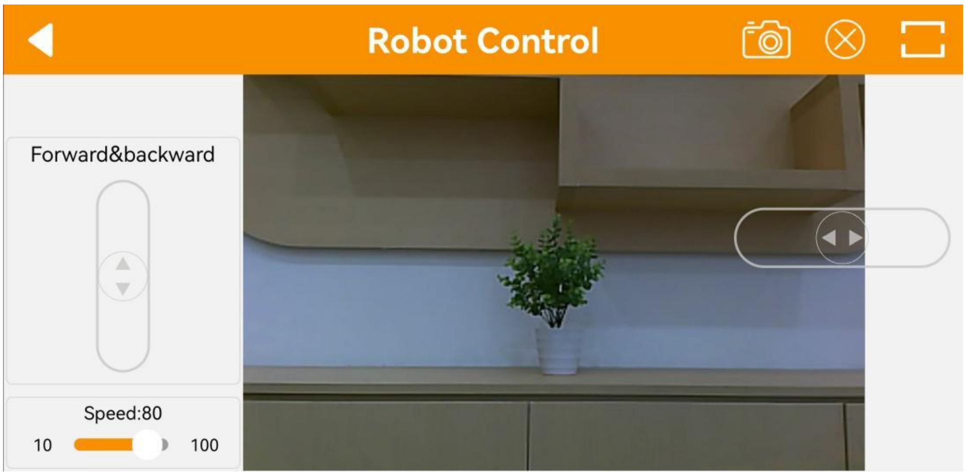
③**Right Controls**:

Control left and right turns.

④Top Menu Bar:

Includes icons for taking screenshots , toggling the navigation bar , and switching to fullscreen mode  (commonly used when operating with a wireless controller).

2) The robot control interface for the Ackerman wheel version is as shown:



① **Left Controls**:

From top to bottom — control forward/backward movement and adjust the speed.

② **Center Display**:

Displays the live camera feed. You can drag the screen to adjust the camera's pan-tilt angle (this feature is available only on the 2D pan-tilt version).

③ **Right Controls**:

Control left and right turns.

④ Top **Menu** Bar:

Includes icons for taking screenshots , toggling the navigation bar , and switching to fullscreen mode  (typically used with a wireless controller).

Note: For the monocular camera version, when the pan-tilt servo reaches its limit and you continue dragging the screen, vibration feedback will occur. This indicates the servo has reached its mechanical limit. Please avoid forcing further rotation, as the pan-tilt has built-in limit protection.

2.2 Lidar

Notes:

① Place the robot in a spacious area to ensure it has enough room to move.

② The Ackermann chassis version does not support the Lidar Guarding function.

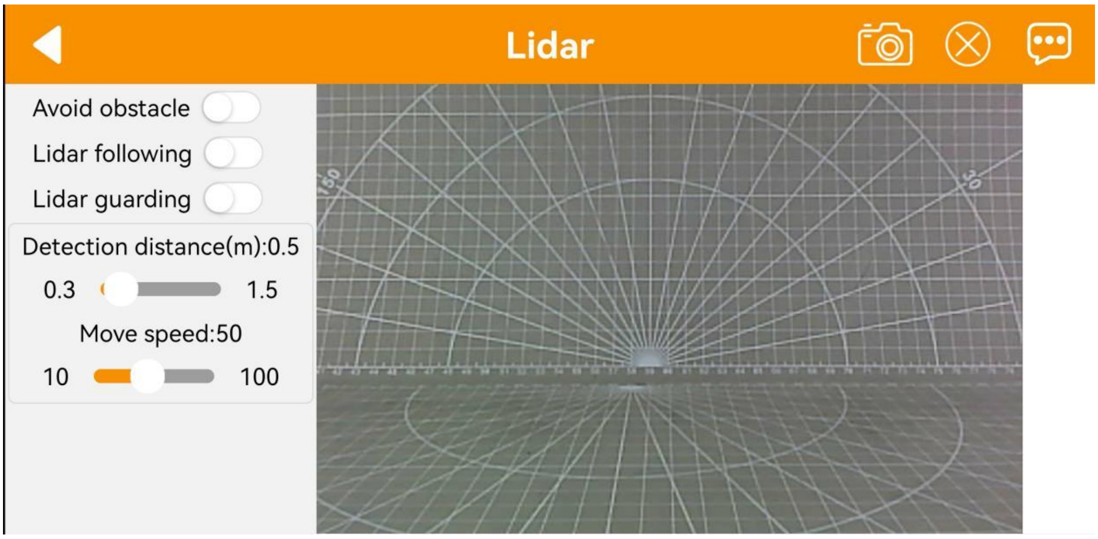
- **Interface introduction:**

The "Lidar" game consists of three modes: avoid obstacle, Lidar following, and Lidar guarding.

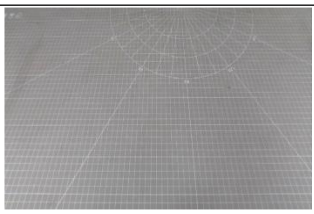
The interface is divided into two parts:

① Left side: switch the games;

② Right side: displays the live camera feed.



- **Function Description**

Icon	Function
Avoid obstacle	Turn on/off the avoid obstacle game.
Lidar following	Turn on/off the Lidar following game.
Lidar guarding	Turn on/off the Lidar guarding game.
	Display the live camera feed.

- **Operation Steps**

① **Avoid Obstacle**

The robot moves forward and, upon detecting an obstacle, will automatically turn to avoid it.

② **Lidar Following**

When an obstacle is detected, the robot adjusts its posture to maintain a specified distance from the obstacle.

③ **Lidar Guarding**

The robot will adjust its orientation to face the obstacle when detected.

2.3 Object Tracking

Notes:

① For optimal tracking performance, it is recommended to tilt the camera slightly downward and move the target object along the ground. This allows the robot to track and follow the object more effectively.

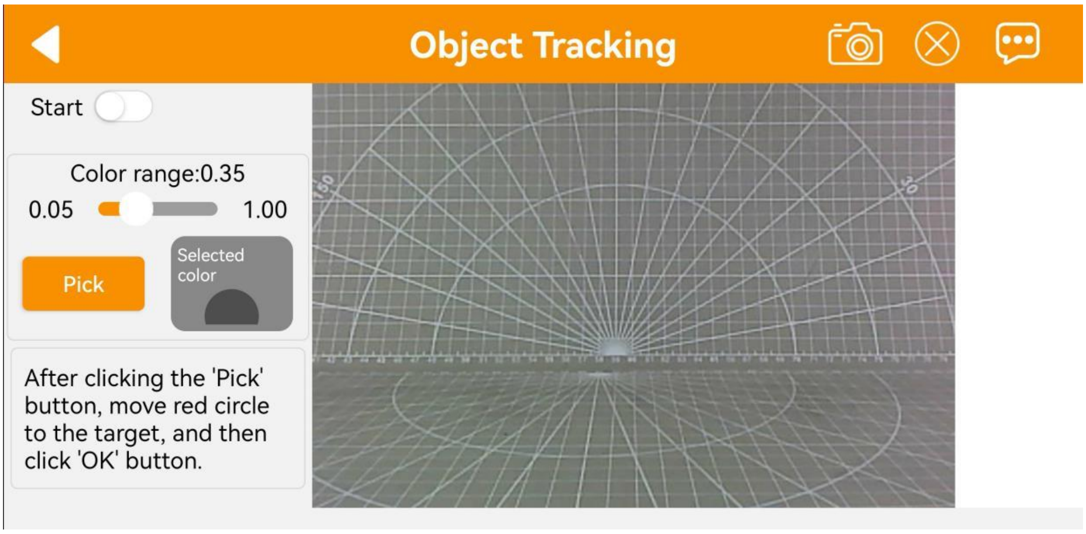
② Carefully adjust the color detection range—avoid setting it too wide or too narrow. A range that is too wide may include unintended colors, while one that is too narrow might prevent the target from being detected. Additionally, ensure that there are no similarly colored objects within the camera's field of view to avoid interference.

- **Interface Introduction**

The interface consists of two parts:

1) Left side: contains the game switch and color picking area;

2) Right side: displays the live camera feed.



- **Function Description**

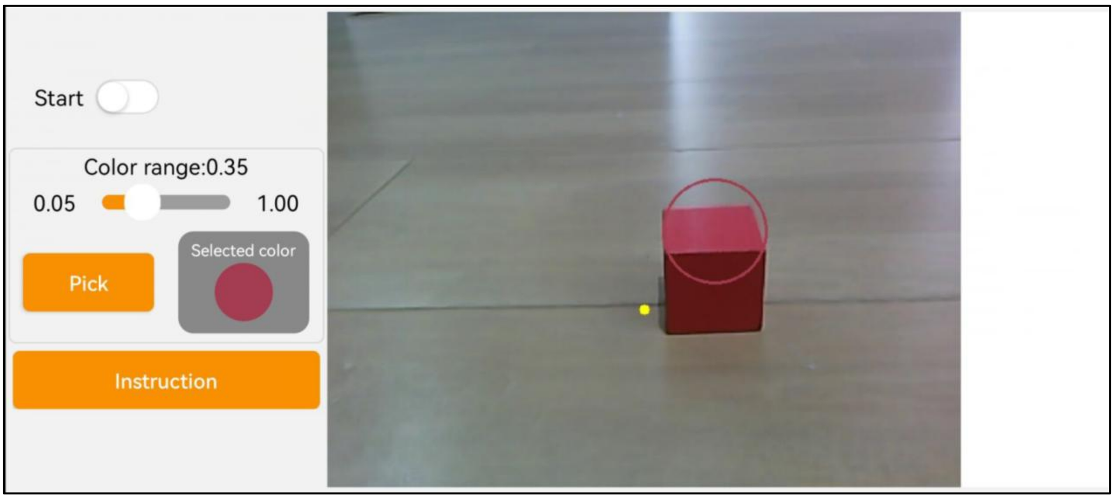
Icon	Function
	Turn the game on or off.
	Adjust the color threshold range (0.05 to 1.00).
	Select a color from the specified area in the live camera feed.
	After clicking "Pick," the button changes to "OK" for confirming the selected color.
	Display the selected color.
	Show the live camera feed.

- **Operation Steps**

1) After tapping the **"Pick"** button, drag the red circle on the live camera feed to the target area to select the color.

2) Once you've tapped the **"OK"** button, the selected color will appear in the box on the right.

3) Tap the **"Start"** button. The robot will then begin tracking and follow the target object as it moves.



2.4 Line Following

Notes:

① Apply tape to the ground to create a path for the robot to follow.

② Adjust the color detection range carefully—avoid setting it too wide or too narrow. A range that is too wide may detect unwanted colors, while a range that is too narrow may fail to recognize the target.

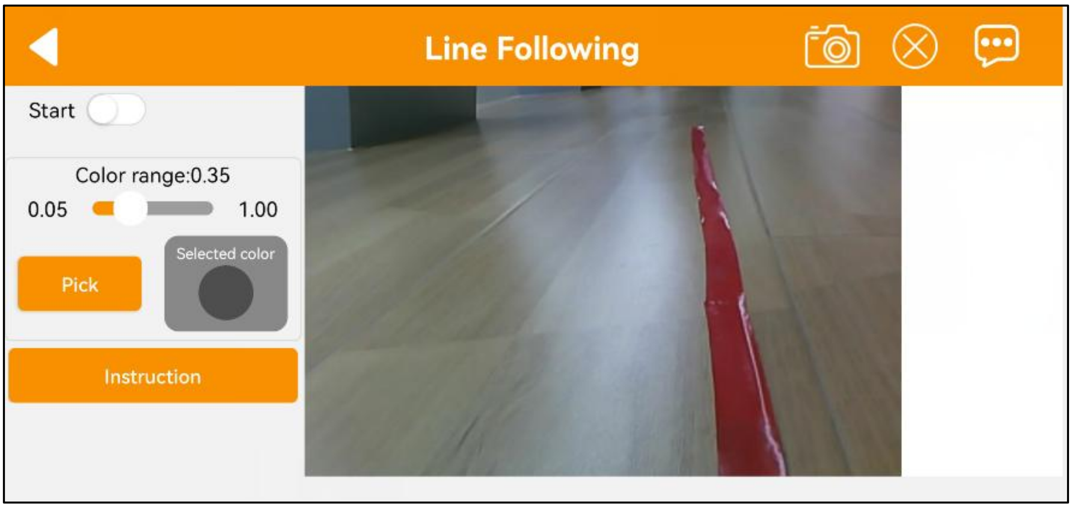
③ Ensure that no objects with colors similar to the target are present within the camera's field of view, as this may affect recognition accuracy.

- **Interface Introduction**

The interface consists of two parts:

1) Left side: contains the game switch and color picking area;

2) Right side: displays the live camera feed.



- **Function Description**

Icon	Function
	Start or stop the game.
	Adjust the color detection threshold (range: 0.05 to 1.00).
	Select a color from a specific area within the live camera feed.
	After tapping "Pick," the button changes to "OK." Tap "OK" to confirm the selected color.
	Display the currently selected color.

	Show the live camera feed.

- **Operation Steps**

1) Tap to enter the game interface. Take red path as example. Click "Pick" button. Drag red circle on live camera feed to the path to pick color, then click "OK" button.

2) The color you pick will display on the right box.

3) Turn on "Start" button, then robot will move along the path.

2.5 Gesture Control

Notes:

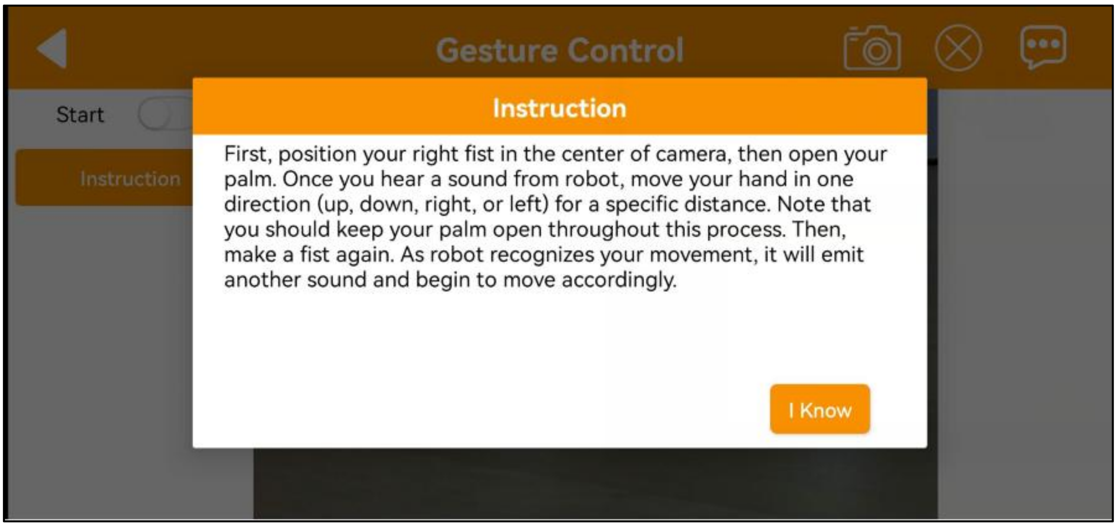
1. This mode only works with your right hand.
2. For proper recognition, your entire palm must be visible on the screen.
3. Move your hand slowly.

- **Interface Introduction**

Click the "Gesture Control" on mode selection interface to enter the game interface. The interface consists of two parts:

1) Left side: contains the game switch and operation instructions;

2) Right side: displays the live camera feed.



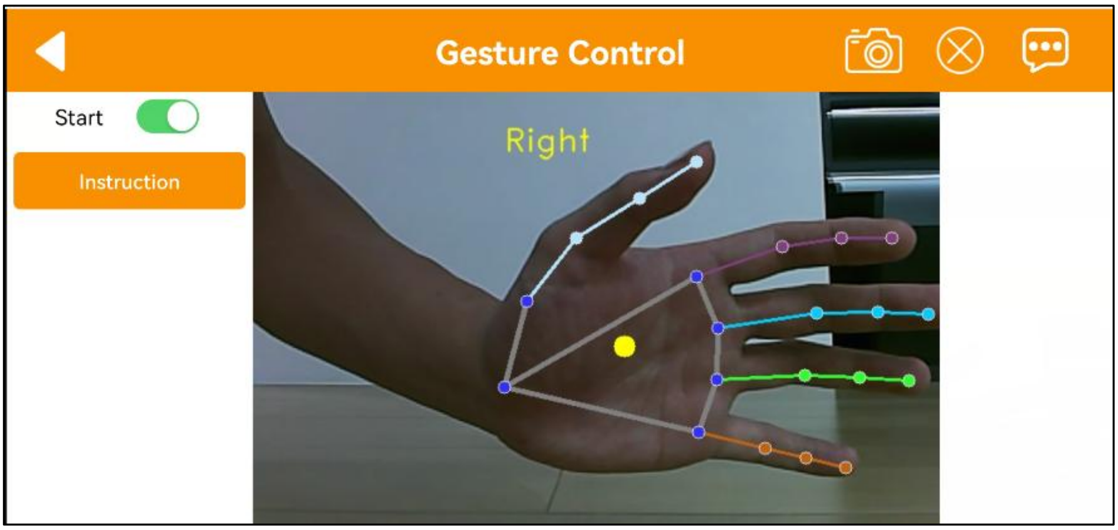
- **Function Description**

Icon	Function
	Turn the gesture control game on or off.
	Display the real-time camera feed. When the palm is open and facing to the right, a trajectory is drawn. When the fist is clenched, the robot will turn right. Draw a trajectory for forward, backward, left, and right movements, and the robot will move

	accordingly—forward, backward, left, or right.

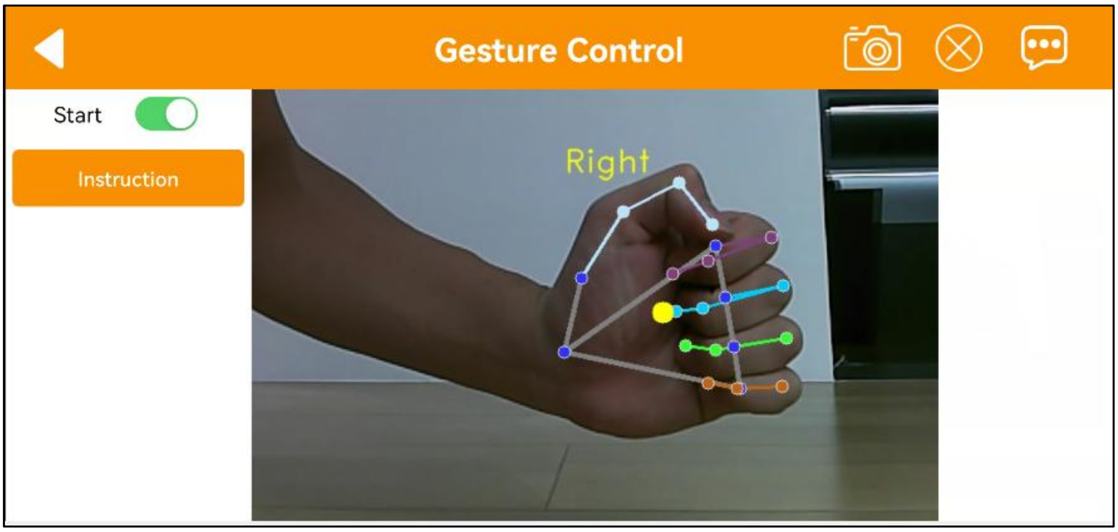
- **Operation Steps**

1) Clench your fist and position your hand at the center of the camera's field of view. Open your fingers and begin moving your hand to draw a trajectory once you hear a beep.



2) While keeping your hand open, move it in one of the following directions—up, down, left, or right—for a certain distance.

3) Make a fist again. After hearing a beep, the robot will start moving in the direction of the drawn trajectory.


Gesture Control
Start
Instruction
Right